

Adrian Ramlal

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RESEARCH FOCUS

Final-semester MAsc researcher (University of Waterloo's Vision and Image Processing Lab) working on physics-grounded neural reconstruction across 3D Gaussian Splatting, differentiable simulation, and reinforcement learning. Eight papers (five first-author), including a Best Imaging Paper award; builds end-to-end PyTorch research systems spanning neural rendering, physics-based simulation, and reinforcement learning, with prior silicon-design and full-stack software engineering. Targeting Research Scientist / Research Engineer roles in computer vision, graphics, and robotics.

EDUCATION

University of Waterloo

Sep. 2024 – Present (Expected Aug. 2026)

Master of Applied Science, Systems Design Engineering

Waterloo, ON

- Thesis: *Physics-Informed Dynamic 3D Reconstruction: From Gaussian Splatting and Rigid-Body Simulation to Inverse Material Estimation*
- **GPA: 97.75%** | Pattern Recognition (100), Computer Graphics (100), Deep Learning (96), Computer Vision (95)
- Advisor: Prof. John S. Zelek

University of Toronto

Sep. 2018 – Jun. 2023

Bachelor of Applied Science, Computer Engineering (Honours)

Toronto, ON

- **GPA: 3.89 / 4.00** | Dean's List (2018–2023) | Double Minor: AI Engineering & Engineering Business
- Capstone: Certificate of Distinction (awarded to top projects in ECE)

RESEARCH EXPERIENCE

Graduate Research Assistant

Sep. 2024 – Present

Vision and Image Processing Lab, University of Waterloo

Waterloo, ON

- Built **PersistGS**, a 4D Gaussian Splatting framework with differentiable rigid-body physics (NVIDIA Newton/Warp) that reconstructs dynamic scenes through full occlusion, raising novel-view PSNR by **+2.46 dB** over kinematic baselines and to within **0.19 dB** of the ground-truth upper bound; modified the simulator's contact kernels to restore gradient flow through collisions, and a centroid-silhouette loss cut trajectory error **40%** versus photometric supervision.
- Built a reinforcement-learning system for inverse material estimation on a non-differentiable MPM fracture simulator, training a neural surrogate on 2,000 simulations ($10^6\times$ speedup) and a goal-conditioned PPO policy in a normalizing-flow latent space that recovers material parameters in 8 evaluations (~ 10 ms, **625 $\times$** fewer than evolutionary search) at **0.642** feature similarity (**0.828** with a warm-start hybrid).
- Integrated 3D Gaussian Splatting into the Genesis physics simulator through a dual-representation system (fixed-topology meshes for collision, Gaussians for rendering), reaching **147.8 FPS** via runtime vertex-buffer updates (**4.65 $\times$** over varying-topology rebuilds); established that physics compatibility cannot be retrofitted post hoc, as fixed-topology conversion incurred **65–80% geometric degradation**, motivating physics-in-the-loop reconstruction.
- Designed a COLMAP-based geometry-aware initialization for static 3DGS, benchmarking five point-cloud upsampling strategies and a depth-guided point-lifting method across 17 scenes; improved reconstruction by **+0.65 dB PSNR** (Replica) and **+0.26 dB** (Mip-NeRF360) and derived scene-characteristic selection guidelines.

Undergraduate Research (Capstone & Project)

Jan. 2021 – Apr. 2023

University of Toronto

Toronto, ON

- Designed an LSTM for controllable music generation conditioned on harmonic and rhythmic parameters, with custom melodic-quality and timbral-similarity metrics folded into the loss; Certificate of Distinction (top ECE projects).
- Built and benchmarked a U-Net-style GAN for greyscale-to-colour translation across medical, historical, and astronomical imagery, comparing perceptual and pixel-level metrics.

PUBLICATIONS

Bold = author; 5 first-author. [Oral] = oral presentation.

- [1] **A. Ramlal** and J. S. Zelek, "PersistGS: Differentiable Physics for Object Permanence in 4D Gaussian Splatting," *CVPR Generative 3D Reconstruction Workshop*, 2026. [Oral] (also presented at ICRA 2026 Real2Sim/Sim2Real Workshop, non-archival)
- [2] **A. Ramlal**, Y. Chen, and J. S. Zelek, "Latent Space Reinforcement Learning for Inverse Material Estimation in Food Fracture Simulation," *CVPR MetaFood Workshop*, 2026. [Poster]

- [3] F. J. Pena Cantu, **A. Ramlal**, J. Lei, V. Agrawal, K. Buzko, Y. Xu, A. Wong, J. S. Zelek, Y. Cui, and Y. Chen, “Office Hours: A Multiday Office Cubicle Dataset for Associative Embodied VQA,” manuscript, 2026.
- [4] **A. Ramlal** and J. S. Zelek, “Real-Time Physics Simulation with Dynamic Mesh-Gaussian Reconstructions,” *JCVIS*, 2025. [Oral]
- [5] **A. Ramlal** and J. S. Zelek, “Beyond Static Gaussians: An Empirical Investigation of Architectural Paradigms for Dynamic 3D Scene Reconstruction,” *JCVIS*, 2025. [Poster]
- [6] S. Jang, **A. Ramlal**, Y. Chen, and P. W. Fieguth, “Anisotropic Kernels for Neural Implicit Surface Reconstruction,” *JCVIS*, 2025. [Oral]
- [7] B. Maser, A. Srebrnjak Yang, **A. Ramlal**, and J. S. Zelek, “Optical Flow-Enhanced Thermal UAV Detection Under Camera Ego-Motion for Real-Time Tactical C-UAV,” *JCVIS*, 2025. [Oral]
- [8] **A. Ramlal**, Y. S. Hu, and J. S. Zelek, “Optimizing 3D Gaussian Splatting via Point Cloud Upsampling,” *JCVIS*, 2024. [Oral, **Best Imaging Paper Award**]

INDUSTRY EXPERIENCE

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| Full-Stack Software Developer | Jul. 2023 – Aug. 2024 |
| <i>Ministry of Education</i> | <i>Port of Spain, Trinidad and Tobago</i> |
| <ul style="list-style-type: none"> Built the role-based access-control and audit-logging layer (granular per-page permissions) securing a national education platform used across the school system. Integrated the arXiv API for automated research-article ingestion and metadata extraction, streamlining content-curation workflows. | |
| Digital Logic Design Intern | May 2021 – Aug. 2022 |
| <i>Intel Corporation</i> | <i>North York, ON</i> |
| <ul style="list-style-type: none"> Authored SystemVerilog behavioural models and testbenches for high-speed multi-protocol serial deserializers across Intel’s chip-to-chip interconnect product lines. Built automation scripts for analog-to-digital schematic translation, cutting manual modelling effort and accelerating design-verification turnaround. Validated mixed-signal model fidelity against silicon measurements with RTL, analog, and verification teams. | |

TEACHING EXPERIENCE

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| Teaching Assistant, Deep Learning & Pattern Recognition (SYDE 577 / 675) | Sep. 2025 – Apr. 2026 |
| <i>University of Waterloo</i> | <i>Waterloo, ON</i> |
| <ul style="list-style-type: none"> Built tutorial materials and interactive Jupyter notebooks on CNNs, transformers, and generative models (SYDE 577); graded work on Bayesian classifiers, kernel methods, and deep architectures and ran weekly graduate office hours (SYDE 675). | |

TECHNICAL SKILLS

Machine Learning & Deep Learning: PyTorch, JAX, TensorFlow, Hugging Face, NumPy, SciPy, scikit-learn, pandas
Computer Vision, Graphics & Simulation: 3DGS, NeRF, COLMAP, OpenCV, OpenGL, Open3D, differentiable rendering (nvdiffrast), differentiable simulation (NVIDIA Warp/Newton), Isaac Lab, Genesis simulator, Blender
Languages: Python, C/C++, CUDA, Rust, MATLAB, SystemVerilog, Bash
Tooling & Infrastructure: Git, Docker, Linux, Google Cloud, Azure, Weights & Biases

AWARDS & FELLOWSHIPS

Engineering Graduate Merit Award , University of Waterloo	2026
Ontario Graduate Fellowship (Engineering) , University of Waterloo	2025
University of Waterloo Graduate Scholarship (first-class average)	2025
International Master’s Award of Excellence , University of Waterloo	2024–2026
Best Imaging Paper Award , JCVIS	2024
David Johnston International Student Entrance Scholarship , University of Waterloo	2024
Engineering International Scholar Award , University of Toronto	2018–2023
Trinidad and Tobago National Open Scholarship (top 1% nationally)	2018
President’s Scholarship of Distinction , University of Waterloo (95%+ admission avg.; declined)	2018
Engineering International Student Scholarship , University of Waterloo (declined)	2018